

Monetary transmission in a fragmented, uncertain world*

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Abstract

This paper investigates the nonlinear relationship between economic uncertainty and the monetary transmission mechanism. We estimate a Threshold Vector Autoregression model on quarterly interest rates, GDP growth, and CPI inflation across a broad range of countries and impose sign restrictions to identify monetary policy shocks. Our findings suggest that monetary shocks lead to larger economic contractions and are followed by more sluggish output adjustments during periods of elevated uncertainty. Efficacy to curb inflation appears to depend on the underlying narrative of uncertainty. Nonetheless, our results support the view that uncertainty acts as a channel that amplifies shocks. For policymakers, this implies that while increased uncertainty does not necessarily block transmission, it raises the costs of reducing inflation.

Keywords: Monetary transmission, Economic uncertainty, Threshold Vector Autoregression, Sign restrictions

JEL: E32, E44, E52, C32

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Introduction

The transmission mechanism of monetary policy and its effects on the economy have long interested academics and policymakers alike. Understanding those dynamics has become even more crucial in an era marked by heightened uncertainty and geoeconomic fragmentation. Uncertainty, whether arising from geopolitical, economic, or policy events can interfere with the transmission channels.

Recent global developments added an additional layer of complexity to modern day economics by introducing the concept of geoeconomic fragmentation. As pointed out by [Ilyina et al. \(2023\)](#), heightened uncertainty levels can result from and contribute to the increasingly fragmented geoeconomic landscape. Fragmentation leaves its imprint on uncertainty about economic policy, geopolitical tensions, and trade. Fortunately all of these have become measurable thanks to advances in analysis of textual data.

Given the extensive literature on uncertainty and the broad consensus about the effects of uncertainty shocks, the implications of heightened uncertainty are relatively well defined and can help shape methodological choices and inference. Yet the interaction between uncertainty and the transmission of monetary policy remains less explored. This gap motivates the central question of this paper: does uncertainty materially affect the transmission mechanism, and if so, what are the policy implications?

With that said, it is important to note the papers, specifically [Aastveit et al. \(2017\)](#) and more recently [Aquino et al. \(2022\)](#), who do address this gap. Their results point to a notable conclusion: episodes of heightened uncertainty do indeed appear to hinder the efficacy of monetary policy intervention. Methodologically, [Aastveit et al. \(2017\)](#) employ an Interacted SVAR in which uncertainty is interacted with interest rates and they define high and low uncertainty at the tails (10th and 90th percentiles) of the distribution. [Aquino et al. \(2022\)](#), by contrast, applies a threshold assumption to the interest-rate equation of their SVAR and uses a grid-search to determine the threshold for regime switches. While we acknowledge the findings of these papers, we argue that a fully interacted system is necessary to properly gauge how the propagation of shocks might differ across regimes. Motivated by these considerations, we develop a parsimonious econometric framework to study the relationship. We estimate a two-regime Threshold Vector Autoregression (TVAR) to examine how dynamics differ across uncertainty regimes. To limit the number

of parameters, the model is deliberately small, including only interest rates, GDP growth, and CPI inflation as endogenous variables. We identify monetary shocks using sign restrictions. We repeat this process across a broad set of countries covering both advanced and emerging market economies.

Compared with earlier findings, our results display meaningful differences. The main difference is in the responses of output to monetary policy shocks. We find that in episodes of high uncertainty the contraction is amplified and more long lasting. This finding is robust across various uncertainty measures, threshold levels and the timing of the regime switches. Whether the uncertainty interferes with the transmission to inflation rates is less clear-cut and appears to depend on the underlying narrative of uncertainty. Overall, we find that uncertainty interferes with the transmission channel in the sense that it increases the cost of curbing inflation.

Why might earlier findings that suggest weaker policy effects under uncertainty be plausible? One strand of the literature broadly agrees that uncertainty shocks have a demand-dampening effect, primarily through reduced investment driven by stronger precautionary saving motives. In such a setting, interest-rate adjustments naturally have less scope to stimulate investment when investment is already depressed. There are, however, plausible counterarguments. Firstly, some literature indicates that in periods of heightened financial stress or recession, monetary policy might be more effective. While high uncertainty and recessions are not necessarily analogous, they share some traits such as reduced demand or risk appetite. Second, high uncertainty can have an amplifying effect on shocks through higher volatility.

The remainder of the paper is organized as follows. The next section reviews the relevant literature. Section three describes the data used in the estimations, with particular emphasis on measures of uncertainty. Section four outlines the methods, including the estimation setup and additional calculations. Section five discusses the results in detail, and section six concludes.

Literature review

Due to having a large pool of literature on related topics, such as uncertainty shocks or the effects of fragmentation, we decided to map out key concepts and implications from previous literature in a compactly summarized format. Below is a table containing such findings and some highlighted papers connected to them:

Highlight of main findings	Highlight of related papers
New uncertainty measures can be derived from textual data.	Baker et al. (2016), Caldara et al. (2020), Caldara and Iacoviello (2022), Ahir et al. (2022)
Uncertainty shocks cause a contraction similar to negative demand shocks. The primary channel is through investments; high uncertainty leads to precautionary savings. Geoeconomic fragmentation is shown to have a severe negative impact on economies.	Bloom (2009), Bloom (2014), Caldara et al. (2016), Cheng and Chiu (2018), Bonciani and Ricci (2020), Nilavongse et al. (2020) Eppinger et al. (2021), Felbermayr et al. (2021), Andriantomanga et al. (2022), Bolhuis et al. (2023) Ilyina et al. (2023)
The IMF Staff Discussion Note points to heightened uncertainty as one channel through which geoeconomic fragmentation affects economies.	
Trade policy uncertainty is shown to reduce investment and export incentives, thereby dampening economic activity.	Sudsawasd and Moore (2006), Osnago et al. (2015), Ebeke and Siminitz (2018), Chen et al. (2021), Wang et al. (2021)
Nonlinear time-series models for causal analysis and forecasting reveal that the impact of shocks is enhanced in recessions, times of high financial stress and uncertainty. Monetary transmission and fiscal multipliers exhibit state dependency.	Avdjiev and Zeng (2014), Schüler (2014), Fry-Mckibbin and Zheng (2016), Colombo et al. (2020), Gbohoui (2021), Schmidt (2020)

Literature on economic uncertainty

Research on uncertainty shocks has already accumulated a substantial body of literature. The interest in this field saw a significant rise following influential advancements in text-based uncertainty measurements ¹ such as Baker et al. (2016) or Caldara and Iacoviello (2022). Several contributions, such as Abel (1983), Bernanke (1983), Rodrik (1991), Bloom et al. (2007), Bloom (2009), Gilchrist et al. (2014), Bloom (2014), Bloom et al. (2018)

¹We give a more thorough insight into such indices and papers that describe how they are constructed in the Data section of this paper.

have reported results consistent with these studies. The common finding is that uncertainty primarily affects the economy by reducing investment, since higher uncertainty encourages precautionary saving.

Baker et al. (2016) show that policy uncertainty shocks contract economic activity by dampening firms' investment and hiring and depressing stock returns. They also document that uncertainty has an amplified effect during downturns and that contractionary uncertainty shocks have a stronger impact than equally sized expansionary ones. Caldara and Iacoviello (2022) present similar findings with respect to geopolitical risks. They link elevated geopolitical risk to lower stock returns, greater market volatility, reduced business investment and ultimately economic contraction. Furthermore, geopolitical risk is also found to have an amplified effect during high financial stress and uncertainty.

Following these seminal papers, many also investigated the impact of uncertainty shocks on US and international data. Caldara et al. (2016) emphasize the role of financial and uncertainty shocks in driving business-cycle fluctuations and argue that the Great Recession likely reflected the interaction of such shocks. Bonciani and Ricci (2020) use local projections to estimate the impact of global financial uncertainty on small open economies, finding contractionary effects that are stronger with greater openness and during downturns.

Colombo (2013) find that US uncertainty shocks spill over substantially to the Euro area business cycle. Furthermore, US shocks are found to contribute more than Euro-area shocks. Nilavongse et al. (2020) estimate a SVAR for the UK with domestic and US policy uncertainty and show that US uncertainty shocks reduce output, whereas domestic shocks play a larger role for exchange rate movements. Biljanovska et al. (2021) report that spillovers can account for roughly two thirds of the contraction from uncertainty shocks, and that policy uncertainty shocks in the US, Europe, and China have the largest effects on Europe and the Western Hemisphere.

Carrière-Swallow and Céspedes (2013) show that uncertainty shocks exert greater influence in emerging economies due to credit constraints. This finding is also supported by Ahir et al. (2019) and Ahir et al. (2022), though Das and Kumar (2018) offer contrasting evidence. Cheng and Chiu (2018) estimate the effects of geopolitical risk shocks on emerging markets and find them to be important drivers of their business cycles. Jung

et al. (2021) use Korean data to show that geopolitical risk triggers sharp declines in stock prices, particularly for large, predominantly domestically owned firms or those with substantial fixed assets.

Literature on geoeconomic fragmentation

Following recent developments, such as the Covid-19 pandemic or the war in Ukraine, fears of deglobalization have been on the rise. Such processes have not been seen in decades, as the world has gradually converged to a globalized economy. This greatly contributes to global uncertainty regarding economic outlooks. Although geoeconomic fragmentation is a more novel terminology, Global Value Chains (GVCs) and trade restrictions have been widely researched, predominantly using general equilibrium models.

Some recent work points to fragmentation causing substantial output losses and added inflationary pressure. Bolhuis et al. (2023) show the adverse effects of fragmentation on output using a new granular dataset, while Andriantomanga et al. (2022) show that supply chain disruptions have a sizable impact on inflation. Both papers conclude the need for policymakers to closely monitor supply chains and adjust policy stances accordingly to mitigate arising disruptions.

Studies on trade restrictions generally show adverse effects for all economies involved. Amiti et al. (2019) and Fajgelbaum et al. (2020) find that protectionist policies are harmful for the US economy in the short run. The former confirms this on a number of other economies as well. Crozet and Hinz (2020) also document a “friendly fire” type effect when Western exporters when sanctioning Russia. Felbermayr et al. (2023) find that doubling non-tariff barriers hurts both the imposing and the targeted economy.

There is broad agreement that GVCs are beneficial and decoupling from them causes welfare losses across the board. Felbermayr et al. (2021) show that real income substantially falls in Europe, should they decouple from China or non-EU economies in the event of a trade war. Heiland et al. (2020) show the benefits of economic integration by estimating the welfare losses from a hypothetical EU disintegration. Dixon et al. (2021) investigates US-China financial decoupling and reports substantial losses in output. While a one-sided limitation of flows from the US to china could boost domestic output, the likely response from China would put the US in a worse position. Banh et al. (2020) and Karpowicz and

Suphaphiphat (2020) show at the firm level that participating in GVCs raises productivity and innovation.

Some research, however, does question the benefits of globalization. Prasad et al. (2007) argue that exposure to GVCs alone need not necessarily spur growth in developing economies. In order to realize gains from global markets, economies first need to have strong institutions and a flexible exchange rate regime. Bergin et al. (2023) find that in some economies capital controls might be beneficial in terms of GDP growth. The arguments are that by being more open to trade and capital flows opening up economies for higher exposure to shocks and global volatility. However, papers such as Eppinger et al. (2021) or D’Aguanno et al. (2021) dismiss this idea by showing that the adverse effects of volatility exposure is either outweighed by the cost of decoupling or that there is no significant exposure in the first place.

How is uncertainty connected to fragmentation? The above body of literature while thoroughly investigating, they miss a crucial aspect in the form of heightened uncertainty. Caldara et al. (2020) and Ahir et al. (2022) both construct measures of trade-policy uncertainty, which can be used to track the uncertainty most closely related to geoeconomic fragmentation over time. Consistent with prior work on uncertainty these papers report that trade policy uncertainty reduces investments. Along with some other papers (e.g., Sudsawasd and Moore (2006), Osnago et al. (2015), Ebeke and Siminitz (2018), Wang et al. (2021), Chen et al. (2021), William and Fengrong (2022)), findings indicate that trade specific uncertainty also reduces exports, lowers risk appetite and drives up firm markups. Besides trade policy uncertainty, the geopolitical risk index of Caldara and Iacoviello (2022) as it captures a variety of events that disrupt global relations.

Literature on nonlinear time series methods

Following the seminal work of Sims (1980), studying business cycles has been dominated with standard linear SVARs. The standard SVAR is an effective tool both in the forecasting and the causal inference literature as it is relatively simple and generally robust. However such linear models have the shortcoming of being unable to asymmetries and state dependency over time. Several nonlinear variations have thus been developed that have seen success forecasting exercises or capturing structural breaks or regime depen-

dence in the propagation of shocks.

Nonlinear models have been used to show the nonlinear nature of uncertainty shocks. Schüler (2014) proposes a quantile SVAR approach to model how uncertainty impacts the economy across the business cycle and finds that uncertainty shocks have a sharper impact in recession periods. Furthermore, the financial system plays an important role in the transmission of uncertainty shocks during recessions. A weak financial sector is shown to be particularly vulnerable to uncertainty shocks, thus deepening the recession. Colombo et al. (2020) use a smooth transition VAR approach to uncover that uncertainty shocks are significantly more severe in recessionary periods versus normal times. Results show that balance sheet tools of central banks are effective at mitigating the impact of uncertainty shocks. Nalban and Smădu (2021) also shows that uncertainty shocks are state dependent. Euro Area aggregates show stronger contraction following uncertainty shocks in times of financial distress. The rebound is also shown to be quicker likely due to monetary policy reaction. Gbohoui (2021) use local projections to show that high uncertainty increases public investment multipliers.

Some papers focus on the state dependent nature of monetary policy as well. Kakes et al. (1998) uses a Markov-Switching model for monetary policy. Results from the US and Germany suggest that monetary policy is more effective in recessionary times. Güney (2018) models the policy functions of the central bank in Turkey and finds asymmetric preferences. The central bank has a more aggressive reaction to inflation gap during recessions, while reaction to growth and inflation uncertainty appears to be more important in expansionary times. Lin (2020) finds asymmetric responses of GDP and investment to interest rate shocks.

A simple and effective tool for state dependence is the Threshold VAR (TVAR). The majority of the literature employs a single threshold (two regimes), such as recessions versus booms or times of high and low financial stress. Some use cases are: Baum and Koester (2011), who show that fiscal multipliers are highly state dependent along the business cycle. Galvão (2006) who TVARs to predict recessions. Alessandri and Mumtaz (2019) who find that the impact of uncertainty shocks depends on financial conditions. Inflationary in normal times, deflationary with larger effect on output during high financial stress.

TVARs have been employed to assess monetary policy as well. Schmidt (2020) shows that monetary shocks have state dependent effect on assets. Macroeconomic risk alters the transmission for risky assets the most, while policy risk matters more for transmission to corporate bonds. Avdjiev and Zeng (2014) finds that the impact of shocks is amplified in times of low growth. Monetary policy is also found to be more aggressive during booms. @li2010financial and Fry-Mckibbin and Zheng (2016) compare monetary policy in high- and low-financial stress states and show that expansions can be effective in high-stress periods. Monetary policy shocks might even move the economy toward low stress. They also document a sharper short-run trade-off between output and inflation in high stress times.

As this section shows, the literature on economic uncertainty and nonlinear dynamics is large and varied. Much of the recent research on monetary policy also suggests state dependence in both policy behavior and the effects of shocks. While many studies tie these state dependent effects to business cycle position or episodes of financial stress, systematic investigation of state-dependent effects driven specifically by uncertainty within monetary-policy research remains relatively limited². This paper aims to fill this gap by examining how uncertainty alters the transmission of monetary shocks by employing multiple measures of uncertainty as well as extending the analysis to a broad set of economies.

Data

Macroeconomic aggregates

To keep the estimations simple, we focus on three core macro indicators: interest rates, GDP, and the CPI. The dataset is quarterly. Interest rates are used in levels, while GDP and CPI enter the model in first differences (quarterly GDP growth and CPI inflation, respectively). We collect quarterly data for a broad set of countries, covering both advanced and emerging market economies. All data are retrieved from the IMF’s database using HAVER analytics. Data is incorporated from the earliest available observation to the second quarter of 2023.

²See the Introduction section for a more details on addressing papers with similar research questions and methods.

Measures of uncertainty

Besides macro aggregates, a key series for our estimations is a measure of uncertainty. Measuring economic uncertainty is not straightforward, and several approaches exist. Historically, uncertainty has been proxied using financial market data. More recently, text-based approaches have been employed to extract information on uncertainty from newspaper articles. Figure 1 shows the uncertainty indices used in this paper. Data for these indices were obtained from policyuncertainty.com.

Baker et al. (2016) construct the Economic Policy Uncertainty (EPU) index from three components: newspaper coverage, tax-code provisions that create uncertainty, and forecaster disagreement. The newspaper component counts articles mentioning policy-uncertainty keywords, the tax-code component captures provisions such as expiring laws or temporary measures, and the forecaster component measures dispersion in economic forecasts. The newspaper component carries the greatest weight in the aggregate index. The authors build a monthly US EPU series from 1985 onward and show it varies substantially over time, spiking around elections, geopolitical events, and major policy changes. Davis (2016) later provides a global version of the EPU index (which is used in this paper), as well as several others created country-specific EPU indices.

Caldara and Iacoviello (2022) propose a newspaper-coverage approach to geopolitical risk (GPR). Their main index uses ten newspapers and, like the EPU, tracks geopolitical risk from 1985. they also construct a Historical GPR (1900–present) based on three newspapers, plus country-specific GPRs for 44 countries. The GPR can be decomposed into eight subcategories (War Threats, Peace Threats, Military Buildups, Nuclear Threats, Terror Threats, Beginning of War, Escalation of War, Terror Acts). To improve accuracy, the authors use unigrams and bigrams (one and two word search terms) along Boolean operators, which is an advance over earlier index constructions such as the EPU.

Ahir et al. (2022) build a quarterly World Uncertainty Index (WUI) from Economist Intelligence Unit reports by measuring the frequency of the word “uncertainty” and its variants in those reports. Several WUI variants exist: country-specific indices for 143 countries, a global average, and group aggregates by region and development. The global WUI tracks major policy shifts and geopolitical events (e.g., 9/11, the Euro-area debt crisis, Brexit). The authors also construct a World Trade Uncertainty Index (WTUI) and World

Uncertainty Spillover Indices (WSUI). The WTUI records occurrences of “uncertainty” near trade-related keywords, while the WSUI records occurrences of “uncertainty” near country-specific keywords (e.g., US, UK, or G7 + China).

Caldara et al. (2020) build three trade-related uncertainty indices. Their primary series uses earnings calls to construct a firm-level Trade Policy Uncertainty index (search terms include “tariff”, “import duty”, “import barrier”). They also create a news-based index (following the EPU methodology) and a tariff-volatility series from a Bayesian stochastic-volatility model. Their results show that firm-level trade-policy uncertainty shocks reduce firm capital—results that are supported by aggregate measures showing contractions in private investment. In this paper we use the news-based Trade Policy Uncertainty (TPU) index because it is publicly available and spans from the 1960s through present day.

PLACEHOLDER FOR FIGURE 1.

To summarize, we use 4 distinct uncertainty measures as threshold variables for the TVAR estimation. The WUI can be considered as the broadest measure of uncertainty, and the only non-news-based index in our roster. Therefore we choose this as our baseline. The EPU captures economic policy related uncertainty, while the GPR and the TPU are the closest measurements we have to capture uncertainty more related to geoeconomic fragmentation.

Methodology

The price puzzle and the use of sign restrictions

An empirical challenge for this analysis is coercing macroeconomic aggregates to behave the way it is written in all macroeconomics textbooks, i.e. to solve the price puzzle. Numerous papers (e.g.: Hanson (2004), Giordani (2004), Demiralp et al. (2014), Bishop et al. (2017), Cloyne and Hürtgen (2016), Romer and Romer (2004)) have discussed the price puzzle. Some solutions include extending models with commodity prices and monetary aggregates, using narrative instruments or turning to sign restrictions. In order to stick to our small scale variable set, the latter option is the best suited for our purposes.

A frequently cited paper in sign restriction literature is the work of Uhlig (2005), introducing a penalty and a rejection algorithm approach. Comparing the two approaches, the rejection algorithm only accepts impulse responses that exactly fit the sign restriction criteria. On the other hand, the penalty algorithm assigns penalty scores to ones that do not exactly fit the sign restriction criteria, and accepts the ones with the lowest penalty scores.

While this flexibility at first might seem favorable, it has some “hidden features” as pointed out by Arias et al. (2018). In our case, as we are working with a full set of restrictions to identify the monetary shock, the artificial introduction of additional (unwanted) restrictions is no issue, however the penalty approach artificially narrows confidence intervals for the estimates, thus giving us a false sense of robustness in the results. For this reason, we implement the full Bayesian rejection algorithm. Rubio-Ramirez et al. (2010) improved upon the efficiency of the estimation process and thus their version of the algorithm is implemented. In all cases, the Bayesian models are estimated with 4 lags and an inverse Wishart prior.

The Threshold VAR approach with sign restrictions

We introduce non-linearity by using each uncertainty index to create an indicator function, taking the interactions of the indicator with each variable, then estimating the VAR model. In essence this means that we assume uncertainty changes the dynamics of the economy. We explore two different threshold levels as indicated in 1, the median and the 70th percentile for each respective index. We acknowledge the fact that a grid-search approach for an optimal threshold level is possible to use in threshold models, however the lack of economic interpretation behind the measurement units of uncertainty indices leads us to instead use threshold levels that refer to positions within their historical distribution.

A generalized mathematical representation of the model would be as follows:

$$Y_t = \sum_{k=1}^4 \Theta_{high} I(X_t \geq \mu) Y_{t-k} + \sum_{k=1}^4 \Theta_{low} I(X_t < \mu) Y_{t-k} + u_t \quad (1)$$

where Y_t is the vector of endogenous variables, Θ_{low} and Θ_{high} are the coefficient matrices for each regime, $I(.)$ is the regime indicator function, X_t is threshold variable, μ is the

threshold value and u_t is the error term.

We would like to note that using contemporaneous values of the threshold variable is unorthodox in the literature of threshold models, as the indicator is most often created based on lagged values. One could argue against our approach that this introduces a confounding bias type effect, as results from the high regime could be interpreted as two shocks impacting the economy simultaneously. However, running the model with changing the specification of Equation 1 above from X_t to X_{t-1} yielded no substantial change in the impulse responses. As the focus of our research is to investigate if interest rate shocks taking place in high uncertainty times have a different impact on the economy compared to low uncertainty times; as well as the results being robust to the changed specification; we believe our approach should be more suited for the purposes of this research.

The vector of endogenous variables Y_t is defined as $Y_t = [r_t, \pi_t, y_t]$, where r_t is the interest rate, π_t is the quarterly inflation rate and y_t is the quarterly GDP growth. Shocks are identified with a full set of sign restrictions for the interest rate shock. We use standard assumptions of a positive monetary policy shock raising interest rates and this subsequently leads to a decline in prices as well as output. As we are only interested in monetary policy shocks, partial identification is sufficient, thus no more restrictions are needed to be specified and only interest rate shocks need to be estimated.

The estimation procedure is done using Markov Chain Monte Carlo (MCMC) simulation with 20000 sampling replications. Impulse responses are constructed following Rubio-Ramirez et al. (2010)³. We begin by drawing reduced form coefficients Θ_i and residuals u_t from the posterior draws. We compute the Cholesky factor P of the residual covariance matrix Σ_u , draw a random orthogonal rotation Q and form $P^* = P Q$ and check whether responses implied by (Θ, P^*) satisfy the sign restrictions on horizons $h = [1, 6]$ repeating procedure up to 200 times per posterior draw. We continue drawing from the posterior until we either reach 1000 accepted impulse responses or all posterior draws have been exhausted.⁴

The above procedure is repeated for each regime, ensuring not only that we have different

³The computations are done in R using the “VARsignR” package created by Danne (2015).

⁴We have no “exact science” reasoning for the choice of 20000 MCMC draws and 200 (sub)draws per MCMC replication. Our goal was to use a number that is large enough to yield a sufficient number of responses that fit the sign restriction criteria, while ensuring that the estimation time would not be so long that it hinders the writing of this paper.

economic dynamics dependent on the uncertainty regime, but also that the contemporaneous effect of the interest rate shock is allowed to be regime specific. The large simulation is necessary in order to find a sufficient number of impulse responses that exactly fit the sign restrictions in each regime. The estimation of the model and shocks is repeated for all countries in the sample and with all uncertainty measurements as indicators.

We would like to acknowledge some limitations of the outlined methodology. Firstly, the setup described above is computationally extensive, necessitating a considerable amount of time to run with every possible specification on every country in our dataset. Secondly, given the large amount of parameters, a number of countries have to be excluded due to an insufficient number of data points; for the remainder of the countries, in a number of cases the algorithm would not find a set of impulse responses that fit the sign restriction criteria; and for a number of countries the simulated impulse responses would be diverging⁵ leading us to exclude them from our final results as well. Initially, we consider 111 countries, however due to the outlined difficulties, we have 36 to 54 countries with valid sets of impulse responses depending on the uncertainty index and threshold level used.

An additional disadvantage of threshold models is their “binary” nature. Although our results demonstrate that higher levels of uncertainty have a drastic effect on how the economy responds to an interest rate shock, our ability to draw conclusions on how the transmission of monetary policy changes over small increments of uncertainty is limited.

Results

Implications from our estimates with the WUI

Having the impulse responses obtained, we proceed to compute the posterior median and an 68% confidence interval for each country-regime combination. Given the large number of countries under examination, we perform another round of aggregation to derive global aggregates, along with several distinct country-group aggregates. These represent the

⁵In each case we consider impulse responses on a 20 quarter horizon. We disregard all estimates where it would be suggested that the effect of the monetary shock by the end of the 20-quarter horizon that exceed a certain limit. We are inclined to discount findings that imply that a unit increase in interest rates would result in a deviation from the steady state of any variable by over 5 percentage points five years post impact. While this is a “rule of thumb” approach, it helps us eliminate diverging outcomes, as well as to make sure we do not overshoot the impact of the shock.

point-by-point medians of the previously calculated posterior median impulse responses and their respective confidence bands. Certain key impulse responses are presented here in the results section, while additional ones are provided in the Annex section of this paper. In all impulse response graphs, we show a 1 percentage point shock to interest rates for the sake of easier comparability.

PLACEHOLDER FOR FIGURE 2.

Aggregating across countries yields insight into the overall implications of elevated uncertainty for interest-rate shocks. Figure 2 displays the response to a 1 percentage point surprise increase in interest rates under each regime, using the WUI as the threshold variable. Looking at inflation, the hypothesis that interest rate shocks become less effective at anchoring inflation under high uncertainty is not supported. The pattern and persistence of inflation responses are broadly similar across regimes; if anything, the initial disinflationary effect appears slightly larger in high uncertainty periods.

Turning to GDP growth, the same shock produces deeper downturns and a more sluggish adjustment of economic activity under high uncertainty. At first glance this might be interpreted as evidence that monetary policy is more effective in recession like states. However, that interpretation is misleading for several reasons. First, any apparent improvement in inflation reduction during high uncertainty episodes is modest. Second, episodes of high uncertainty do not fully coincide with recessions. Finally, as the larger output contraction is accompanied with similar inflation responses, the correct interpretation would be that high uncertainty raises the output tradeoff for curbing inflation.

Addressing different uncertainty measures

The responses across countries exhibit strong heterogeneity which may not be apparent from the previous figure. Moreover, the “narrative”⁶ of uncertainty also matters for the responses under each regime. In Figure 3 we address these differences by reporting impulse responses across all indices, as well as comparing the responses of Small and Large Economies⁷.

⁶Or in other words, which uncertainty index we use as a threshold variabel.

⁷To further investigate, we repeat the aggregation by replacing the country groups with Advanced Economies and Emerging Economies in place of Large and Small Economies, as it can be seen in Figure

PLACEHOLDER FOR FIGURE 3.

Response patterns across the board are consistent with our earlier estimates for all aggregates. Output contraction is larger conditional on being in the high uncertainty regime in both groups of economies irrespective of which uncertainty measure is used. Transmission to inflation rates for the most part appear to be unaffected, especially in large economies and broadly speaking, interest rates adjust similarly as well. In general, uncertainty appears to distort the transmission channel of small economies much more than for large economies, as a mostly unchanged transmission to inflation is accompanied by a markedly larger drop in activity, when the interest rate shock happens in high uncertainty times.

There are some responses that are not in agreement with the majority of the results. For instance, the response of inflation in high GPR times appears to be weaker in small economies. On the other hand, for large economies, the transmission to either aggregate seems to be unaffected by high GPR. Comparatively, the transmission channel appears to be more affected by high EPU for advanced economies. Output shows a much larger contraction compared with other indices albeit this is accompanied by a larger drop in inflation as well⁸.

These results seem to be in line with what we could expect. EPU plays a stronger role in influencing the transmission mechanism in large economies. This makes sense, as the index most likely picks up on economic policy events happening in larger economies. From this perspective, it is closer in interpretation to what high local uncertainty would exert on interest rate shock transmission. The fact that small economies show a similar responses in high EPU is supported by their general vulnerability to international shocks and contagion. The resilience to international disruptions in large economies, and high exposure in small economies is also supported by how high GPR shapes the impulse responses.

The response patterns when using either the TPU or the WUI appear to be quite similar.

⁷ in the Annex. The key implications do not change, however the difference between the country groups is less apparent. This leads us to believe that the size of the country matters more than its level of development.

⁸One might argue that there is some endogeneity issues at play with the EPU index and monetary policy shocks, as the search queries of the EPU index do contain terms such as “central bank”, “federal reserve” or “interest rate”. To what extent might this be possible on a global scale we are unsure. We do however provide a short analysis in the Annex on US data and find no sign of policy shocks being correlated with the Global EPU index.

The WUI picks up on uncertainty whether or not its source is a local or global event and trade related uncertainty can disrupt in large economies as well. Broadly, these responses support the idea that uncertainty distorts the transmission channel by increasing the cost of reducing inflation. With respect to transmission to inflation, contemporaneous responses appear larger in high uncertainty times for the most part, which would support the idea that uncertainty increases volatility. Lastly, the responses with high GPR and EPU do suggest that to some extent the narrative of uncertainty matters with respect to how uncertainty distorts the transmission channel.

Addressing the level of uncertainty

The next logical question is if uncertainty matters for the transmission mechanism is how much does an increase in uncertainty influence it. While the binary nature of the threshold model does not allow for such analysis, we can shift the threshold value to gain some insights. In an attempt to investigate this, we raise the threshold value from the median to the 70th percentile and rerun the models and compare the results with the ones obtained before. Figure 4 reports these results using the WUI index for the regime switches⁹.

PLACEHOLDER FOR FIGURE 4.

Raising the threshold from median to 70th percentile resulted GDP and inflation responses shifting slightly down in both regimes. While these downward shifts are minor, and were we to report the confidence bands, one likely could not tell apart the two sets of responses, we can still make some conclusions. Firstly, shifting the definition of high uncertainty from the median to a more extreme value of the distribution shifted both responses downwards. This likely means that higher uncertainty leads to monetary shocks contracting the economy even more, even if that increment is minor. Moving on to the low uncertainty responses, them shifting downwards as well indicates that the the median level is already disruptive for the transmission of monetary policy shocks. As the response of inflation shifted down

⁹To further investigate, we ran the models with every specification. We also created a set of comparisons for large, small, advanced and emerging economies with the two threshold levels, all of which can be seen in Figures 8, 9, 10 and 11. Results from these remain consistent with the main graphs supporting our findings.

in both regimes, we are inclined to conclude that uncertainty serves as a conduit that amplifies shocks, like some previous research has found as well.

Implications for policymakers and researchers

Our results on inflation are inconclusive: whether uncertainty directly undermines monetary policy’s ability to anchor inflation depends on the uncertainty index used. Apart from an initial drop, inflation responses show only small differences between low- and high-uncertainty regimes. By contrast, the effects on output are clear and robust across all indices: monetary shocks produce a sharper and more persistent contraction during periods of high uncertainty. The slower recovery is especially pronounced in large economies, while small economies tend to experience a deeper immediate fall in activity.

Our findings appear to be consistent with the idea that high uncertainty amplifies the impact of shocks. However, this amplification is not symmetric across the propagation of different aggregates. Put differently, high uncertainty need not break the monetary transmission mechanism, but it raises the cost of disinflation. Recognizing this trade-off should encourage central banks to be more deliberate when tightening policy in uncertain times. Our results also point to the value of alternative tools: managing expectations via clear forward guidance or other communication strategies can help “cut through the fog” of uncertainty and reduce the real costs of bringing inflation down.

Conclusion

In this paper we investigate the relationship between uncertainty and how effective the monetary transmission is. We employ small scale TVARs and identify monetary policy shocks with sign restrictions across a wide range of economies. Evidence on whether uncertainty directly makes central banks less effective at anchoring inflation is mostly inconclusive. On the other hand, we find robust evidence that monetary shocks produce a larger contraction in real activity in times of high uncertainty. These together lead us to the conclusion that uncertainty is disruptive for central banks in the sense that it raises the trade-off between disinflation and growth.

Dynamics appear to differ across economies. Generally the transmission in large economies is more resilient to global uncertainty compared to small open economies. For large economies, economic policy uncertainty seems to be the most disruptive and geopolitical risks the least. Quite contrary, monetary transmission in small economies suffers the most from high geopolitical risks. Higher uncertainty also appears to be marginally more harmful, as a more extreme subset along the uncertainty distribution showed slightly larger contractions. Lastly, our findings indicate that the median level of the uncertainty is disruptive enough to yield meaningfully different output responses between the two regimes.

Even though we ran a number of specifications to prove the robustness of our results, a number of research questions still remain open. While we checked if higher levels of uncertainty affect the transmission, our results are with respect to this are rather simple. Such an investigation would require moving beyond the two-regime TVAR framework to either one with more regimes, a smooth transition variation, or using a time-varying-parameter framework. We also leave unexplored whether contractionary or expansionary shocks show more or less pronounced sign asymmetry under high uncertainty as well as whether global or local measures of uncertainty matter more for policymakers.

References

- Aastveit, Knut Are, Gisle James Natvik, and Sergio Sola. 2017. "Economic Uncertainty and the Influence of Monetary Policy." *Journal of International Money and Finance* 76: 50–67.
- Abel, Andrew B. 1983. "Optimal Investment Under Uncertainty." *The American Economic Review* 73 (1): 228–33.
- Ahir, Hites, Nicholas Bloom, and Davide Furceri. 2019. "Caution: Trade Uncertainty Is Rising and Can Harm the Global Economy." *VoxEU. Org* 4.
- . 2022. "The World Uncertainty Index." National bureau of economic research.
- Alessandri, Piergiorgio, and Haroon Mumtaz. 2019. "Financial Regimes and Uncertainty Shocks." *Journal of Monetary Economics* 101: 31–46.
- Amiti, Mary, Stephen J Redding, and David E Weinstein. 2019. "The Impact of the 2018 Tariffs on Prices and Welfare." *Journal of Economic Perspectives* 33 (4): 187–210.
- Andriantomanga, Zo, Marijn Bolhuis, and Shushanik Hakobyan. 2022. "Global Supply Chain Disruptions: Challenges for Inflation and Monetary Policy in Sub-Saharan Africa."
- Aquino, Juan Carlos, Nelson R Ramirez-Rondán, and Luis Yépez Salazar. 2022. "Does Uncertainty Matter for the Effectiveness of Monetary Policy?"
- Arias, Jonas E, Juan F Rubio-Ramírez, and Daniel F Waggoner. 2018. "Inference Based on Structural Vector Autoregressions Identified with Sign and Zero Restrictions: Theory and Applications." *Econometrica* 86 (2): 685–720.
- Avdjiev, Stefan, and Zheng Zeng. 2014. "Credit Growth, Monetary Policy and Economic Activity in a Three-Regime TVAR Model." *Applied Economics* 46 (24): 2936–51.
- Baker, Scott R, Nicholas Bloom, and Steven J Davis. 2016. "Measuring Economic Policy Uncertainty." *The Quarterly Journal of Economics* 131 (4): 1593–1636.
- Banh, Hang T, Mr Philippe Wingender, and Cheikh A Gueye. 2020. *Global Value Chains and Productivity: Micro Evidence from Estonia*. International Monetary Fund.
- Baum, Anja, and Gerrit Koester. 2011. "The Impact of Fiscal Policy on Economic Activity over the Business Cycle-Evidence from a Threshold VAR Analysis."
- Bergin, Paul, Woo Jin Choi, and Ju H Pyun. 2023. "Catching up by 'Deglobalizing': Capital Account Policy and Economic Growth." National Bureau of Economic Research.
- Bernanke, Ben S. 1983. "Irreversibility, Uncertainty, and Cyclical Investment." *The Quarterly Journal of Economics* 98 (1): 85–106.
- Biljanovska, Nina, Francesco Grigoli, and Martina Hengge. 2021. "Fear Thy Neighbor: Spillovers from Economic Policy Uncertainty." *Review of International Economics* 29 (2): 409–38.
- Bishop, James, Peter Tulip, et al. 2017. *Anticipatory Monetary Policy and the "Price Puzzle"*. Reserve Bank of Australia.
- Bloom, Nicholas. 2009. "The Impact of Uncertainty Shocks." *Econometrica* 77 (3): 623–85.
- . 2014. "Fluctuations in Uncertainty." *Journal of Economic Perspectives* 28 (2): 153–76.
- Bloom, Nicholas, Stephen Bond, and John Van Reenen. 2007. "Uncertainty and Investment Dynamics." *The Review of Economic Studies* 74 (2): 391–415.
- Bloom, Nicholas, Max Floetotto, Nir Jaimovich, Itay Saporta-Eksten, and Stephen J Terry. 2018. "Really Uncertain Business Cycles." *Econometrica* 86 (3): 1031–65.
- Bolhuis, Marijn, Jiaqian Chen, and Benjamin Kett. 2023. "Fragmentation in Global Trade: Accounting for Commodities."

- Boncianni, Dario, and Martino Ricci. 2020. "The International Effects of Global Financial Uncertainty Shocks." *Journal of International Money and Finance* 109: 102236.
- Caldara, Dario, Cristina Fuentes-Albero, Simon Gilchrist, and Egon Zakrajšek. 2016. "The Macroeconomic Impact of Financial and Uncertainty Shocks." *European Economic Review* 88: 185–207.
- Caldara, Dario, and Matteo Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review* 112 (4): 1194–1225.
- Caldara, Dario, Matteo Iacoviello, Patrick Molligo, Andrea Prestipino, and Andrea Raffo. 2020. "The Economic Effects of Trade Policy Uncertainty." *Journal of Monetary Economics* 109: 38–59.
- Carrière-Swallow, Yan, and Luis Felipe Céspedes. 2013. "The Impact of Uncertainty Shocks in Emerging Economies." *Journal of International Economics* 90 (2): 316–25.
- Chen, Tao, Huasheng Gao, and Yuxi Wang. 2021. "Tariff Uncertainty and Firm Innovation: Evidence from the US–China Permanent Normal Trade Relation." *Journal of Empirical Finance* 62: 12–27.
- Cheng, Chak Hung Jack, and Ching-Wai Jeremy Chiu. 2018. "How Important Are Global Geopolitical Risks to Emerging Countries?" *International Economics* 156: 305–25.
- Cloyne, James, and Patrick Hürtgen. 2016. "The Macroeconomic Effects of Monetary Policy: A New Measure for the United Kingdom." *American Economic Journal: Macroeconomics* 8 (4): 75–102.
- Colombo, Valentina. 2013. "Economic Policy Uncertainty in the US: Does It Matter for the Euro Area?" *Economics Letters* 121 (1): 39–42.
- Colombo, Valentina, Alessia Paccagnini, et al. 2020. *The Asymmetric Effects of Uncertainty Shocks*. Australian National University, Crawford School of Public Policy, Centre for
- Crozet, Matthieu, and Julian Hinz. 2020. "Friendly Fire: The Trade Impact of the Russia Sanctions and Counter-Sanctions." *Economic Policy* 35 (101): 97–146.
- D'Aguanno, Lucio, Oliver Davies, Aydan Dogan, Rebecca Freeman, Simon Lloyd, Dennis Reinhardt, Rana Sajedi, and Robert Zymek. 2021. "Global Value Chains, Volatility and Safe Openness: Is Trade a Double-Edged Sword?" *Bank of England Financial Stability Paper*, no. 46.
- Danne, Christian. 2015. *VARsignR: Estimating VARs Using Sign Restrictions*.
- Das, Debojyoti, and Surya Bhushan Kumar. 2018. "International Economic Policy Uncertainty and Stock Prices Revisited: Multiple and Partial Wavelet Approach." *Economics Letters* 164: 100–108.
- Davis, Steven J. 2016. "An Index of Global Economic Policy Uncertainty." National Bureau of Economic Research.
- Demiralp, Selva, Kevin D Hoover, and Stephen J Perez. 2014. "Still Puzzling: Evaluating the Price Puzzle in an Empirically Identified Structural Vector Autoregression." *Empirical Economics* 46: 701–31.
- Dixon, Peter, James A Giesecke, Jason Nassios, and Maureen T Rimmer. 2021. "Finance in a Global CGE Model: The Effects of Financial Decoupling Between the US and China." *Journal of Global Economic Analysis* 6 (2).
- Ebeke, Mr Christian H, and Jesse Siminitz. 2018. *Trade Uncertainty and Investment in the Euro Area*. International Monetary Fund.
- Eppinger, Peter, Gabriel J Felbermayr, Oliver Krebs, and Bohdan Kukharskyy. 2021. "Decoupling Global Value Chains."
- Fajgelbaum, Pablo D, Pinelopi K Goldberg, Patrick J Kennedy, and Amit K Khandelwal.

2020. “The Return to Protectionism.” *The Quarterly Journal of Economics* 135 (1): 1–55.
- Felbermayr, Gabriel, Steffen Gans, Hendrik Mahlkow, and Alexander-Nikolai Sandkamp. 2021. “Decoupling Europe.” Kiel Policy Brief.
- Felbermayr, Gabriel, Hendrik Mahlkow, and Alexander Sandkamp. 2023. “Cutting Through the Value Chain: The Long-Run Effects of Decoupling the East from the West.” *Empirica* 50 (1): 75–108.
- Fry-McKibbin, Renée, and Jasmine Zheng. 2016. “Effects of the US Monetary Policy Shocks During Financial Crises—a Threshold Vector Autoregression Approach.” *Applied Economics* 48 (59): 5802–23.
- Galvão, Ana Beatriz C. 2006. “Structural Break Threshold VARs for Predicting US Recessions Using the Spread.” *Journal of Applied Econometrics* 21 (4): 463–87.
- Gbohoui, William. 2021. *Uncertainty and Public Investment Multipliers: The Role of Economic Confidence*. International Monetary Fund.
- Gilchrist, Simon, Jae W Sim, and Egon Zakrajšek. 2014. “Uncertainty, Financial Frictions, and Investment Dynamics.” National Bureau of Economic Research.
- Giordani, Paolo. 2004. “An Alternative Explanation of the Price Puzzle.” *Journal of Monetary Economics* 51 (6): 1271–96.
- Güney, Pelin Öge. 2018. “Asymmetries in Monetary Policy Reaction Function and the Role of Uncertainties: The Case of Turkey.” *Economic Research-Ekonomska Istraživanja* 31 (1): 1367–81.
- Hanson, Michael S. 2004. “The ‘Price Puzzle’ Reconsidered.” *Journal of Monetary Economics* 51 (7): 1385–1413.
- Heiland, Inga, Gabriel Felbermayr, and Jasmin Groeschl. 2020. “Complex Europe: Quantifying the Cost of Disintegration.” Discussion Paper 15200, CEPR.
- Ilyina, Ms Anna, Mr Shekhar Aiyar, Mr Jiaqian Chen, Mr Christian H Ebeke, Mr Roberto Garcia-Saltos, Tryggvi Gudmundsson, Mr Alvar Kangur, et al. 2023. “Geo-Economic Fragmentation and the Future of Multilateralism.”
- Jung, Seungho, Jongmin Lee, and Seohyun Lee. 2021. *Geopolitical Risk on Stock Returns: Evidence from Inter-Korea Geopolitics*. International Monetary Fund.
- Kakes, Jan et al. 1998. *Monetary Transmission and Business Cycle Asymmetry*. Graduate School/Research Institute Systems, Organisation; Management.
- Karpowicz, Ms Izabela, and Mrs Nujin Suphaphiphat. 2020. *Productivity Growth and Value Chains in Four European Countries*. International Monetary Fund.
- Li, F, and P St-Amant. 2010. “Financial Stress, Monetary Policy, and Economic Activity. Bank of Canada.” Working Paper. Go to original source.
- Lin, Tzu-Yu. 2020. “Asymmetric Effects of Monetary Policy.” *The BE Journal of Macroeconomics* 21 (2): 425–47.
- Nalban, Valeriu, and Andra Smădu. 2021. “Asymmetric Effects of Uncertainty Shocks: Normal Times and Financial Disruptions Are Different.” *Journal of Macroeconomics* 69: 103331.
- Nilavongse, Rachatar, Rubaszek Michał, and Gazi Salah Uddin. 2020. “Economic Policy Uncertainty Shocks, Economic Activity, and Exchange Rate Adjustments.” *Economics Letters* 186: 108765.
- Osnago, Alberto, Roberta Piermartini, and Nadia Rocha. 2015. “Trade Policy Uncertainty as Barrier to Trade.” WTO Staff Working Paper.
- Prasad, Eswar S, Kenneth Rogoff, Shang-Jin Wei, and M Ayhan Kose. 2007. “Financial Globalization, Growth and Volatility in Developing Countries.” In *Globalization and*

- Poverty*, 457–516. University of Chicago Press.
- Rodrik, Dani. 1991. “Policy Uncertainty and Private Investment in Developing Countries.” *Journal of Development Economics* 36 (2): 229–42.
- Romer, Christina D, and David H Romer. 2004. “A New Measure of Monetary Shocks: Derivation and Implications.” *American Economic Review* 94 (4): 1055–84.
- Rubio-Ramirez, Juan F, Daniel F Waggoner, and Tao Zha. 2010. “Structural Vector Autoregressions: Theory of Identification and Algorithms for Inference.” *The Review of Economic Studies* 77 (2): 665–96.
- Schmidt, Jörg. 2020. “Risk, Asset Pricing and Monetary Policy Transmission in Europe: Evidence from a Threshold-VAR Approach.” *Journal of International Money and Finance* 109: 102235.
- Schüler, Yves S. 2014. “Asymmetric Effects of Uncertainty over the Business Cycle: A Quantile Structural Vector Autoregressive Approach.”
- Sims, Christopher A. 1980. “Macroeconomics and Reality.” *Econometrica: Journal of the Econometric Society*, 1–48.
- Sudsawasdi, Sasatra, and Robert E Moore. 2006. “Investment Under Trade Policy Uncertainty: An Empirical Investigation.” *Review of International Economics* 14 (2): 316–29.
- Uhlig, Harald. 2005. “What Are the Effects of Monetary Policy on Output? Results from an Agnostic Identification Procedure.” *Journal of Monetary Economics* 52 (2): 381–419.
- Wang, Hui, Huayu Shen, Xiaoyi Tang, Zuofeng Wu, and Shuming Ma. 2021. “Trade Policy Uncertainty and Firm Risk Taking.” *Economic Analysis and Policy* 70: 351–64.
- William, Mbanye, and Wang Fengrong. 2022. “Economic Policy Uncertainty and Industry Innovation: Cross Country Evidence.” *The Quarterly Review of Economics and Finance* 84: 208–28.
- Wu, Jing Cynthia, and Fan Dora Xia. 2016. “Measuring the Macroeconomic Impact of Monetary Policy at the Zero Lower Bound.” *Journal of Money, Credit and Banking* 48 (2-3): 253–91.

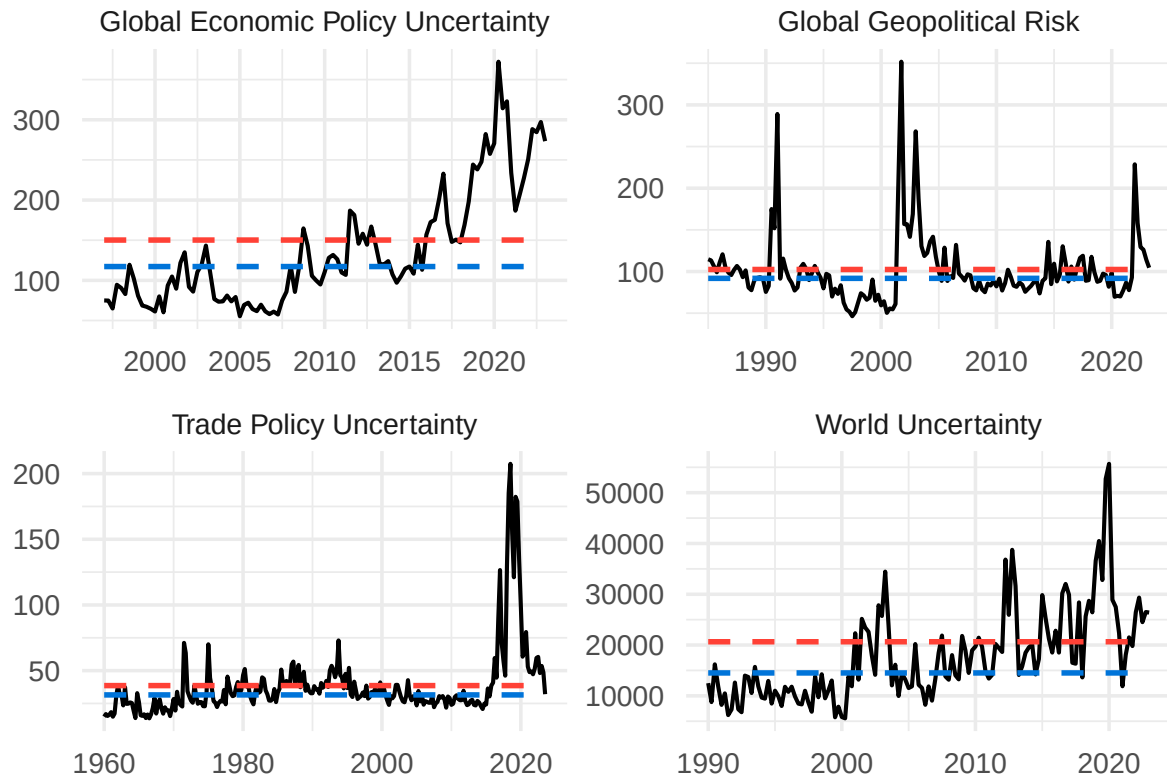


Figure 1: Historical data of uncertainty measures.

Notes: The blue line indicates the median, while the red line indicates the 70th percentile of the respective index.

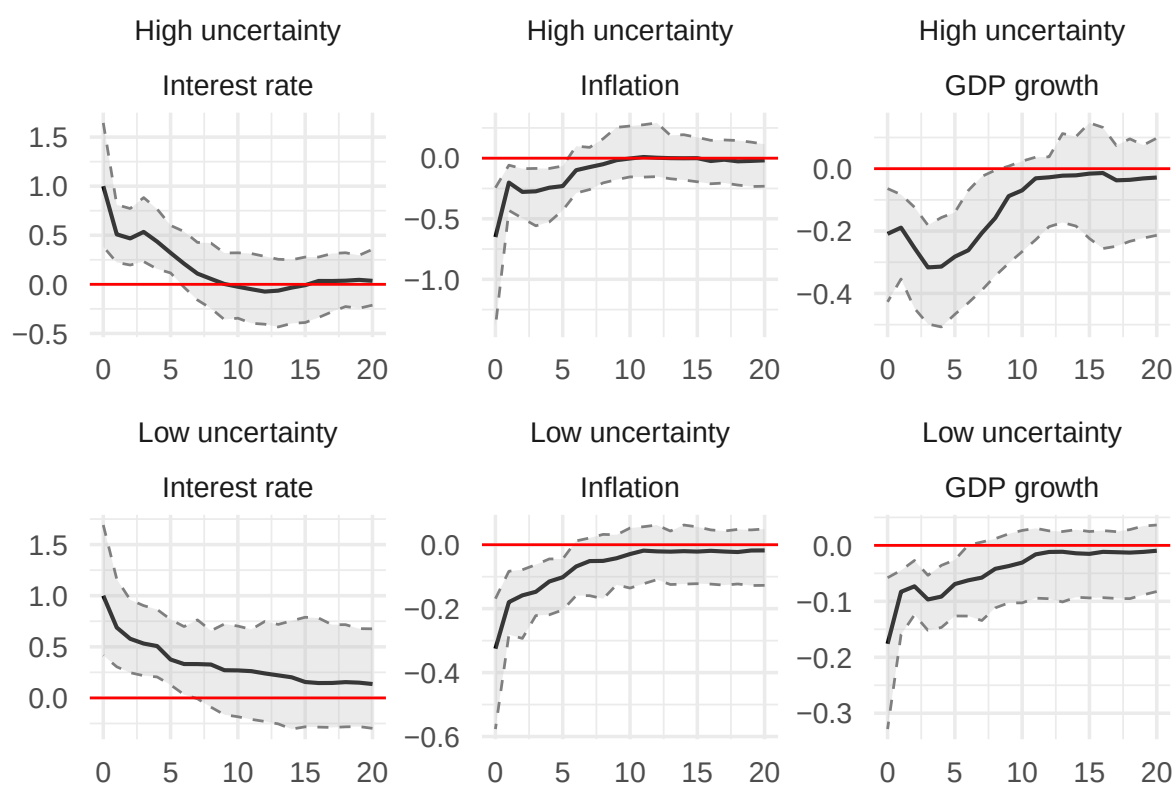


Figure 2: Global IRFs with the WUI and median threshold.

Notes: Point-by-point median impulse responses aggregating over all countries with valid impulse responses. The World Uncertainty Index is used as the threshold variable with the threshold level set to the median. The confidence intervals are the 68% confidence level.

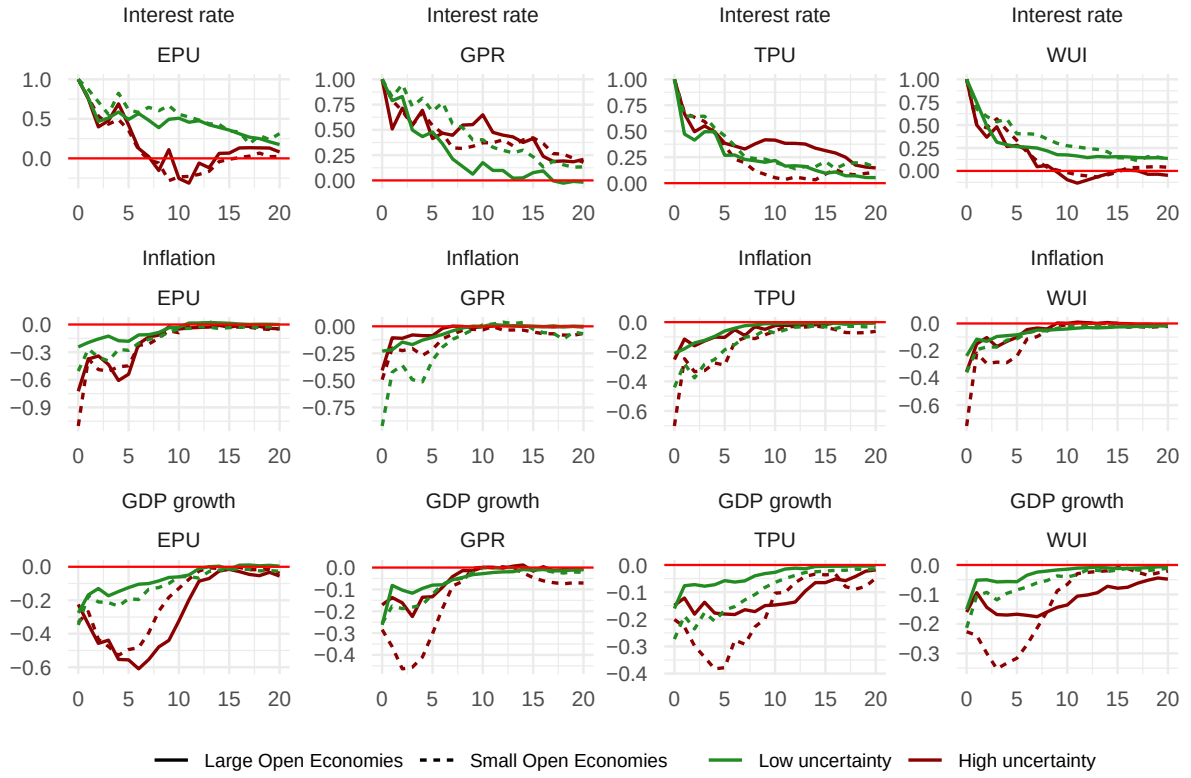


Figure 3: Small and Large Economy IRFs, median threshold.

Notes: Point-by-point median impulse responses comparing Small and Large Economies with valid impulse responses. The threshold value for each uncertainty index is set to its median.

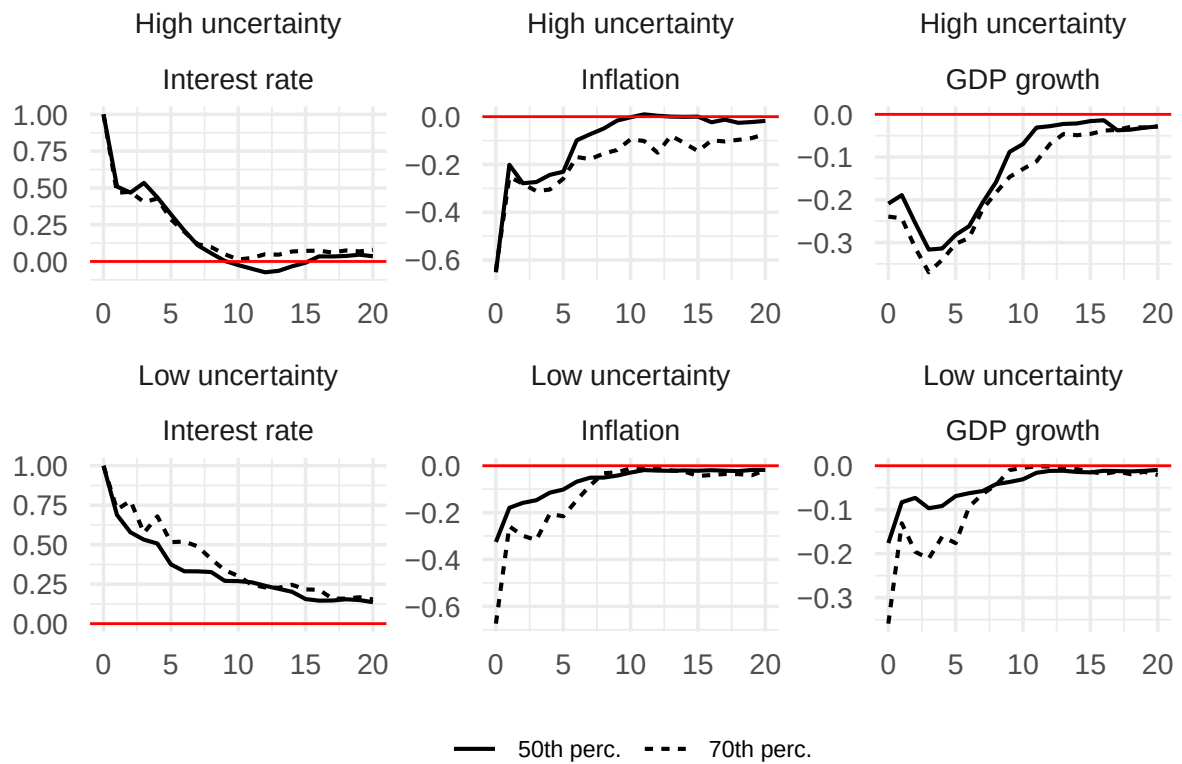


Figure 4: Global IRFs with the WUI, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in the whole sample with valid impulse responses. The World Uncertainty Index is used as the threshold variable.

Annex

Annex A - Further evidence from US data

This section of the Annex intends to further our investigation, by changing from a birds-eye view to a more microscopic perspective focusing on implications for the United States. This allows us to carry out more meaningful robustness checks, as a richer set of data is available for the US. Specifically, in this section we are testing for three things: i) does adding extra controls - the uncertainty index and a financial stress measure - change the results?; ii) alternative interest rates for estimating the policy shock - this allows us to use the Wu and Xia (2016) Shadow Rate to check if the presence of a Zero-Lower-Bound biases our results; and iii) we check if there are any endogeneity concerns between the estimated policy shock series and the uncertainty index. To keep this section concise, we only report results obtained with the Baker et al. (2016) EPU index as the threshold variable.

Controlling for uncertainty and financial stress: In our original setup, we considered a parsimonious 3-equation TVAR model with interest rates, inflation and GDP growth, using an exogenous threshold variable. While this setup is effective in investigating our research question - does economic uncertainty change the monetary transmission mechanism - it leaves some questions unanswered. In Baker et al. (2016) (and subsequent additions to the literature) it was shown that economic uncertainty is an important driver of business cycles. Naturally, this raises the question: does not accounting for uncertainty introduce some omitted variable bias that substantially changes our results?

Moreover, economic uncertainty is often associated with high financial stress. Li and St-Amant (2010) and Fry-McKibbin and Zheng (2016) show that financial stress interferes with the transmission of monetary policy, and interventions in times of high stress can often be more effective. Could this be behind our results? To check the robustness of our results to a potential omitted variable bias, we consider three more model specifications:

$$Y_t = \sum_{k=1}^4 \Theta_{high} I(X_t \geq \mu) Y_{t-k} + \sum_{k=1}^4 \Theta_{low} I(X_t < \mu) Y_{t-k} + \sum_{k=1}^4 \beta X_{t-k} + \epsilon_t \quad (2)$$

$$Y_t = \sum_{k=1}^4 \Theta_{high} I(X_t \geq \mu) Y_{t-k} + \sum_{k=1}^4 \Theta_{low} I(X_t < \mu) Y_{t-k} + \sum_{k=1}^4 \gamma Z_{t-k} + \epsilon_t \quad (3)$$

$$Y_t = \sum_{k=1}^4 \Theta_{high} I(X_t \geq \mu) Y_{t-k} + \sum_{k=1}^4 \Theta_{low} I(X_t < \mu) Y_{t-k} + \sum_{k=1}^4 \beta X_{t-k} + \sum_{k=1}^4 \gamma Z_{t-k} + \epsilon_t \quad (4)$$

where X_{t-k} represents the inclusion of four lags of the uncertainty indicator, and Z_{t-k} represent the inclusion of four lags of the St. Louis Fed Financial Stress Index. The graph below reports results obtained with these specifications along with the results of our baseline estimate for the US.

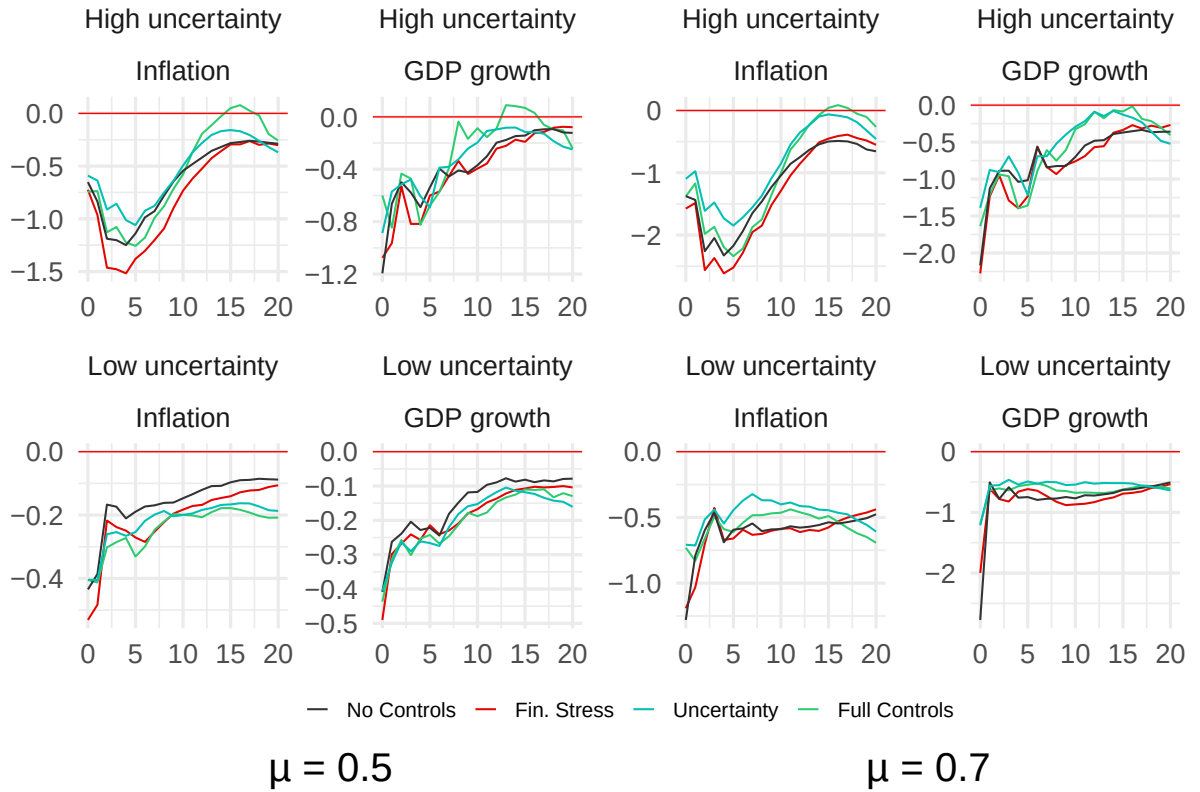


Figure 5: Robustness check across different control variable structures using US data.

Through visual inspection of the graphs, we can draw the following conclusions. Responses aside from high uncertainty inflation stay relatively similar across either specification.

When controlling for uncertainty, the transmission of policy shocks becomes slightly more muted. This should be in line with the uncertainty literature, as economic uncertainty can be a significant driver of fluctuations. On the other hand, when controlling for financial stress, the responses slightly sharpen. This again would be in line with literature, as policy shocks during high uncertainty times can calm the markets, thus here we are accounting for another transmission channel. Accounting for both, makes the responses essentially makes the responses indistinguishable from the baseline model for the first 10 quarters following the shock. Overall, the differences are relatively minor, and the same implications can be drawn irrespective of which model specification we consider. While this exercise did uncover some evidence for an omitted variable bias, this should not be too concerning.

Robustness across different interest rates: When investigating the efficacy of monetary policy shocks in post-GFC times, academics and policymakers alike tend to question if the ZLB interferes with results obtained with an empirical model. Without an appropriate measure to track implied interest rates, this question can be difficult to tackle. When focusing on the US alone, however, the Wu and Xia (2016) Shadow Rate can serve as an appropriate measure to overcome this difficulty. In order to check if the ZLB interferes with our results, we estimate the baseline model again, replacing the Fed Funds Rate with the Shadow Rate between early 2009 through 2016 and 2020 through 2022. Additionally, we estimate the model another time, using the return on 1-year Treasury Bills, as longer yields are also less susceptible to the ZLB.

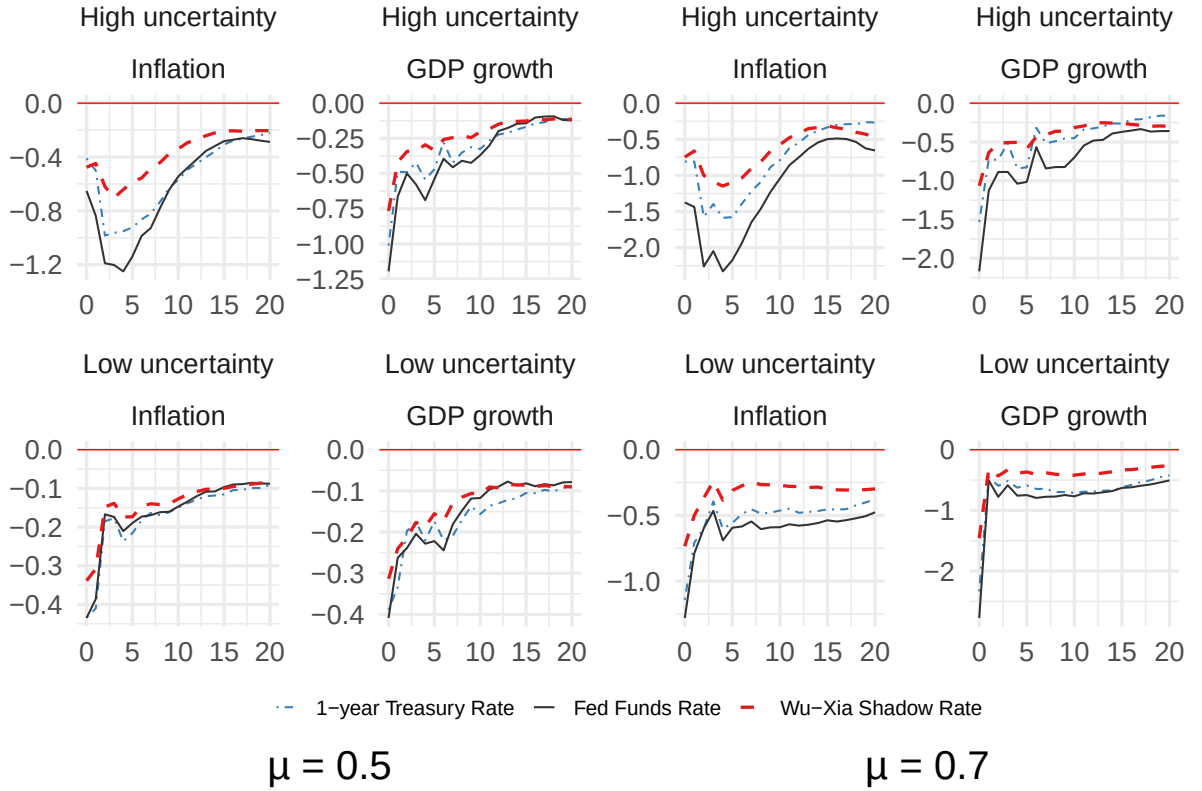


Figure 6: Robustness check across different interest rates using US data.

When using the Shadow Rate, aside from low uncertainty responses with $\mu = 0.5$ (i.e. the threshold value set to the median), the IRFs shift noticeably upward. This points to the fact that indeed the correct specification should be one that properly accounts for the existence of the ZLB. Nevertheless, regardless of whether or not we use an interest rate that is adjusted for the ZLB or not, we arrive at the same conclusions. Additionally,

on a broader scale, a number of less advanced economies in our sample did not face a ZLB period, making this slightly less of a severe concern. Additionally, the Shadow Rate is seldom available for countries, responses obtained with a slightly longer yield fail to properly account for the ZLB, and the above mentioned reasons, we are not too concerned by this with respect to the robustness of our results.

Endogeneity: Studying the relationship between policy shocks and (policy) uncertainty naturally raises the question of whether or not the uncertainty index is endogenous to policy innovations. To test for this, we run Granger causality tests between the EPU index and the posterior median of the residuals obtained from various model specifications. We summarize a select few of these here in Table 2.

Model	Test 1	Test 2
1 Linear	0.4	0.055
2 Base TVAR, Median	0.705	0.246
3 Base TVAR, 70th perc.	0.917	0.621
4 Full control TVAR, Median	0.813	0.288
5 Full control TVAR, 70th perc.	0.978	0.32
6 Full control TVAR with shadow rate, Median	0.484	0.362
7 Full control TVAR with shadow rate, 70th perc.	0.98	0.325

Table 2: Granger causality tests with alternative model specifications

Notes:

Test 1: H_0 : Shock does not cause uncertainty

Test 2: H_0 : Uncertainty does not cause shock

The results of these tests suggests that in most cases we fail to reject the null, and thus policy shocks do not drive uncertainty, neither the other way around. In fact, all non-linear specifications outperform the standard linear SVAR. Interpreting the p-values as the likelihood of H_0 to be true, in a linear setting, there is a high likelihood of uncertainty Granger-causing the monetary shock. As the p-values of the test stay relatively similar across specifications, we are not concerned that endogeneity between the policy shock series and policy uncertainty would exist. Furthermore, this exercise reinforces our belief that the threshold specification is superior in this context.

Annex B - Additional graphs referenced in the paper

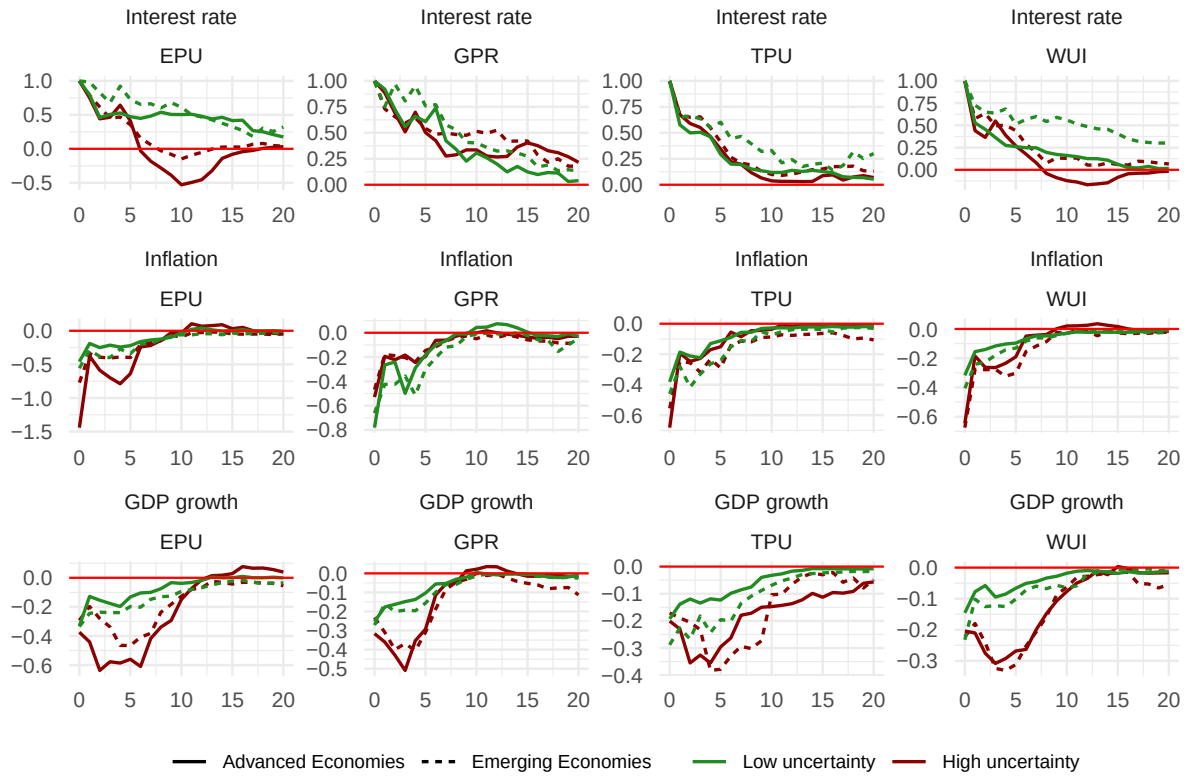


Figure 7: Advanced and Emerging Economy IRFs, median threshold.

Notes: Point-by-point median impulse responses comparing Advanced and Emerging with valid impulse responses. The threshold value for each uncertainty index is set to its median.

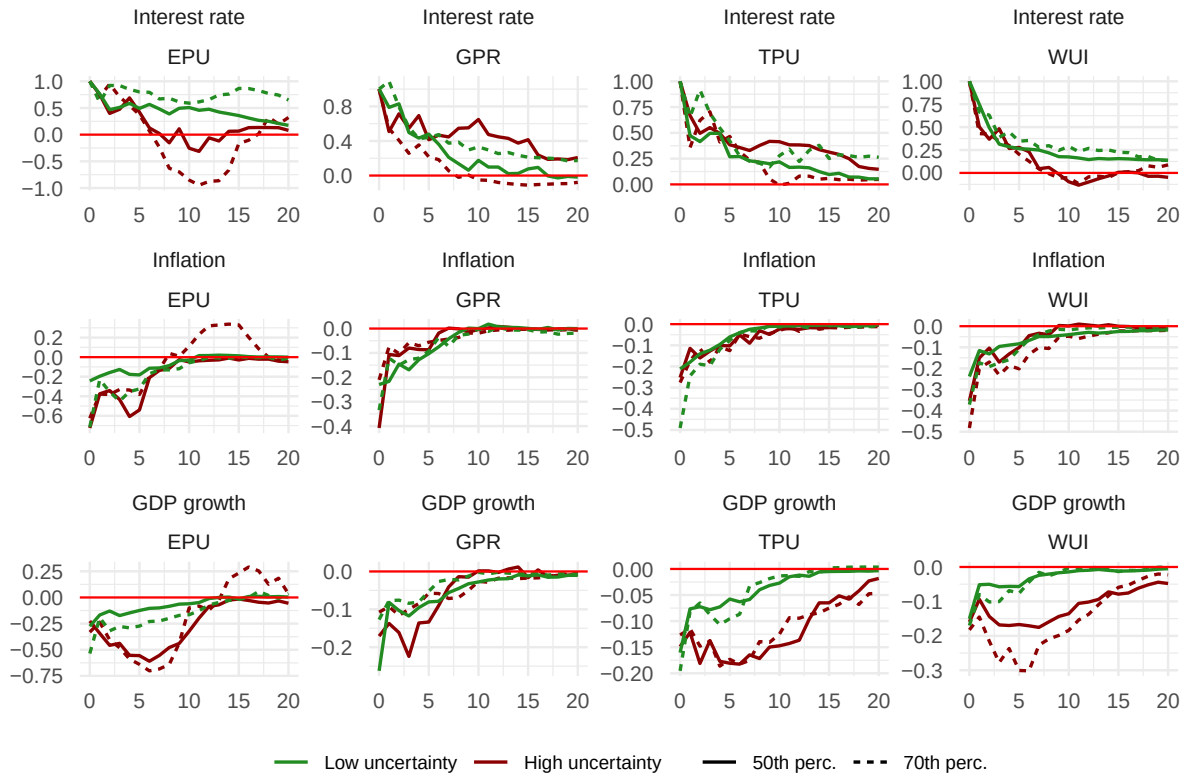


Figure 8: Large Economy IRFs, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in Large Open Economies with valid impulse responses.

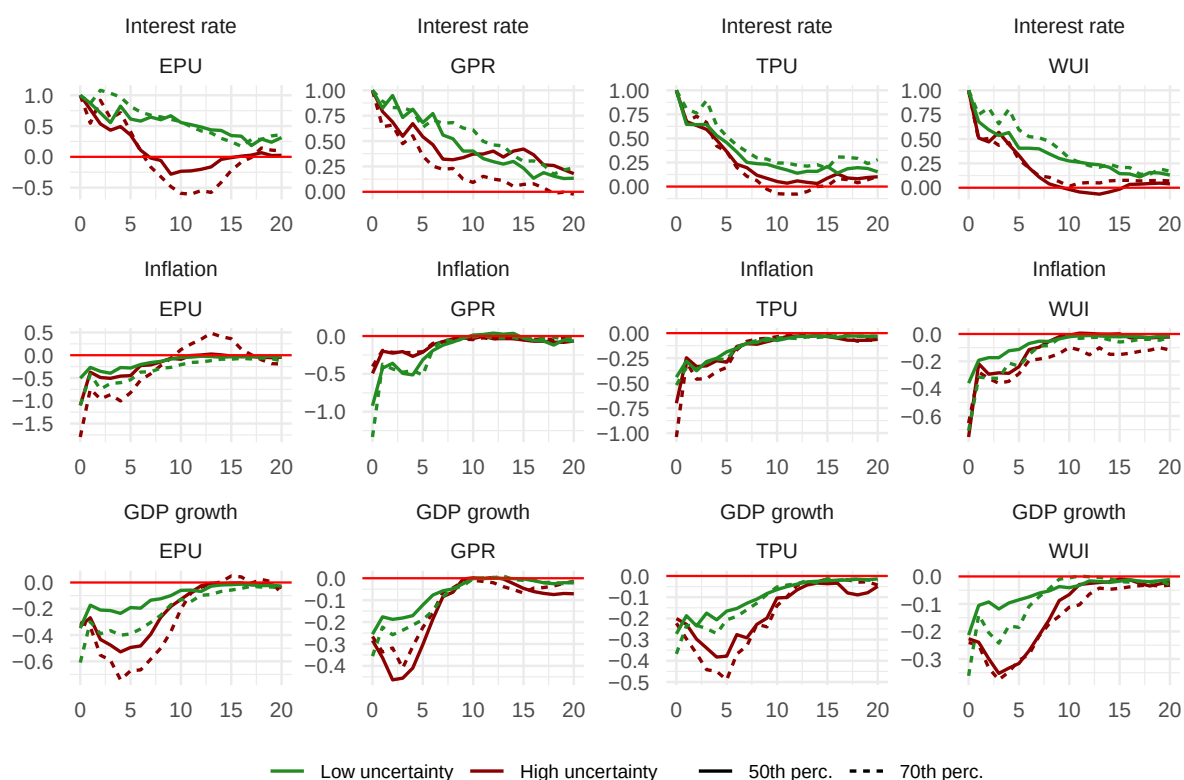


Figure 9: Small Economy IRFs, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in Small Open Economies with valid impulse responses.

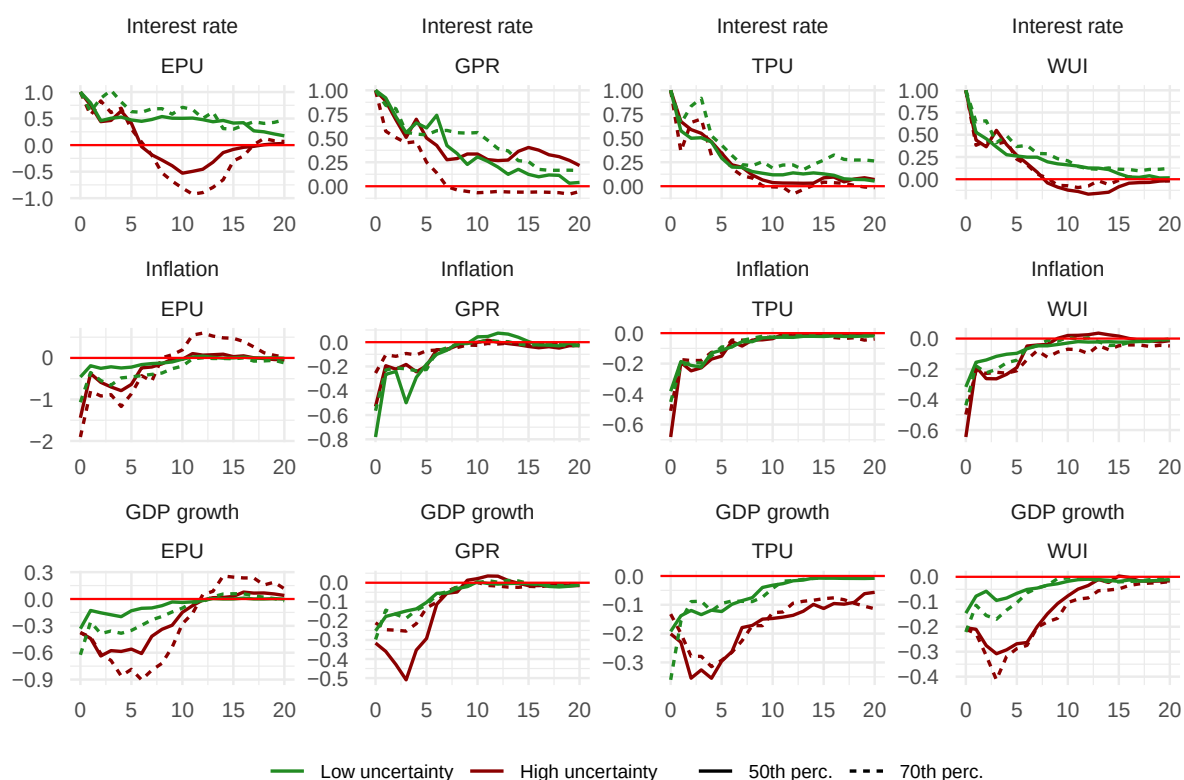


Figure 10: Advanced Economy IRFs, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in Advanced Economies with valid impulse responses.

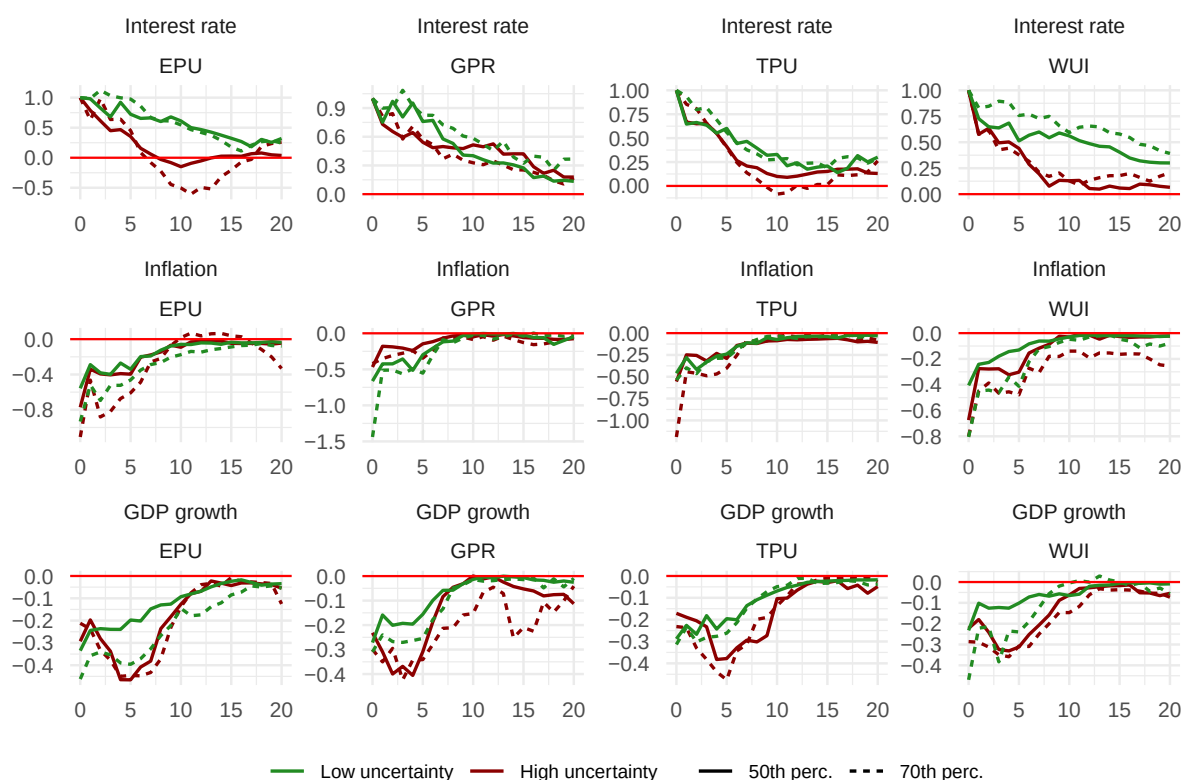


Figure 11: Emerging Economy IRFs, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in Emerging Economies with valid impulse responses.

Annex C - Variants of Figures 2 and 4

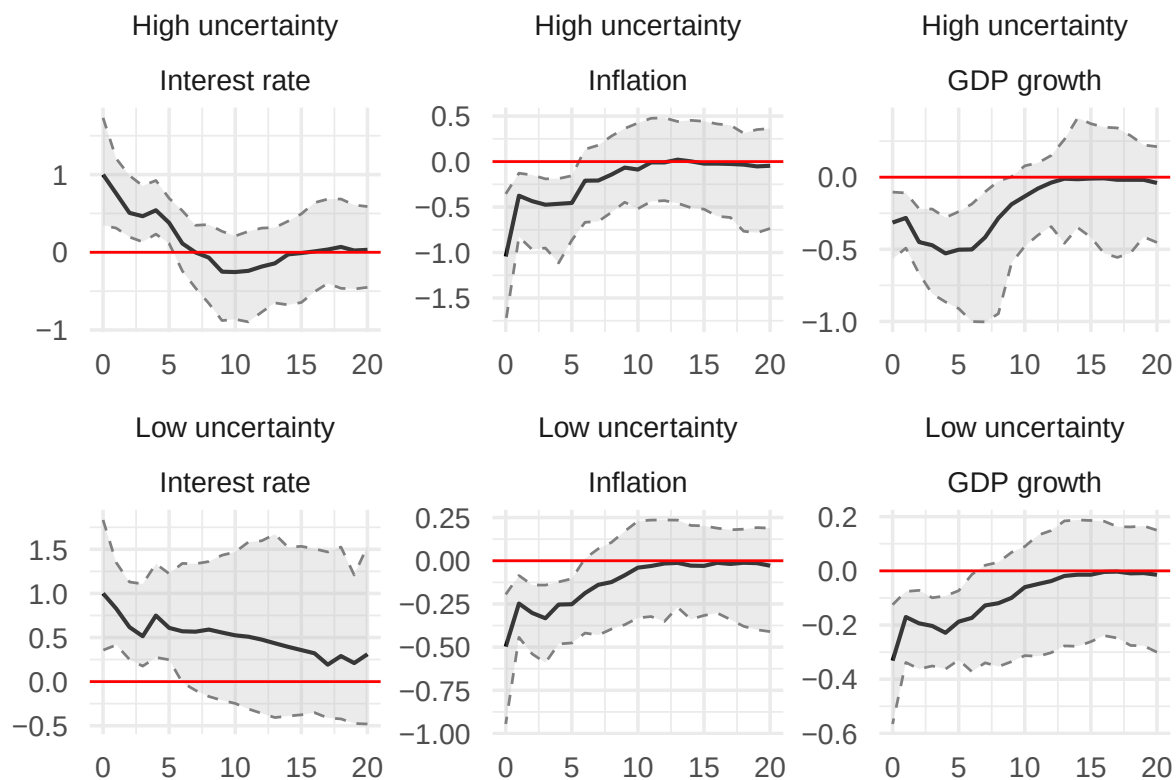


Figure 12: Global IRFs with the EPU and median threshold.

Notes: Point-by-point median impulse responses aggregating over all countries with valid impulse responses. The Economic Policy Uncertainty Index is used as the threshold variable with the threshold level set to the median. The confidence intervals are the 68% confidence level.

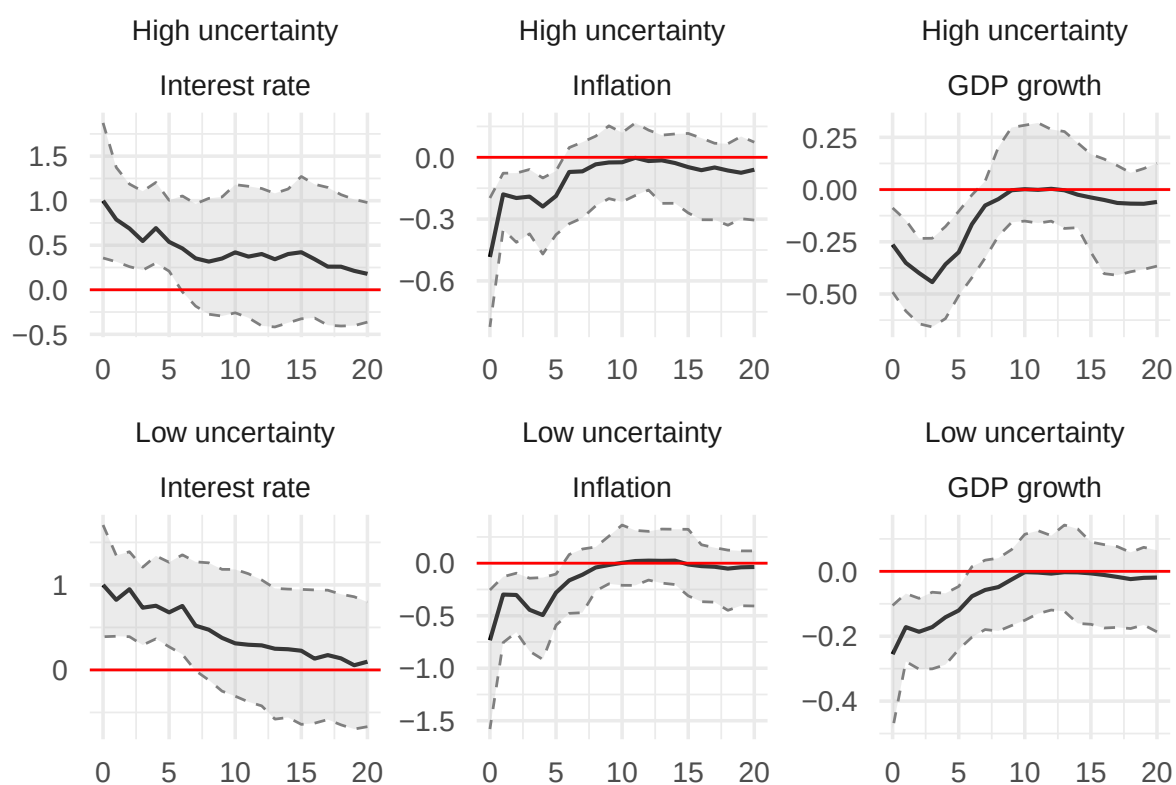


Figure 13: Global IRFs with the GPR and median threshold.

Notes: Point-by-point median impulse responses aggregating over all countries with valid impulse responses. The Geopolitical Risk Index is used as the threshold variable with the threshold level set to the median. The confidence intervals are the 68% confidence level.

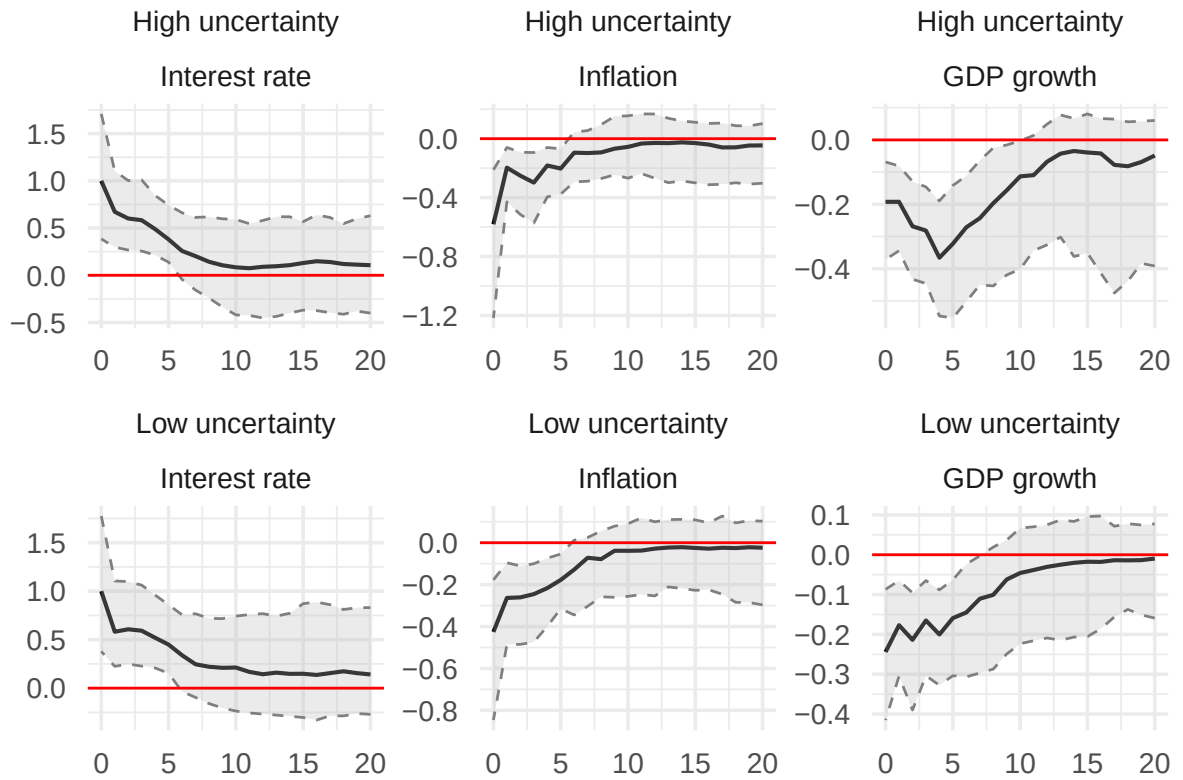


Figure 14: Global IRFs with the TPU and median threshold.

Notes: Point-by-point median impulse responses aggregating over all countries with valid impulse responses. The Trade Policy Uncertainty Index is used as the threshold variable with the threshold level set to the median. The confidence intervals are the 68% confidence level.

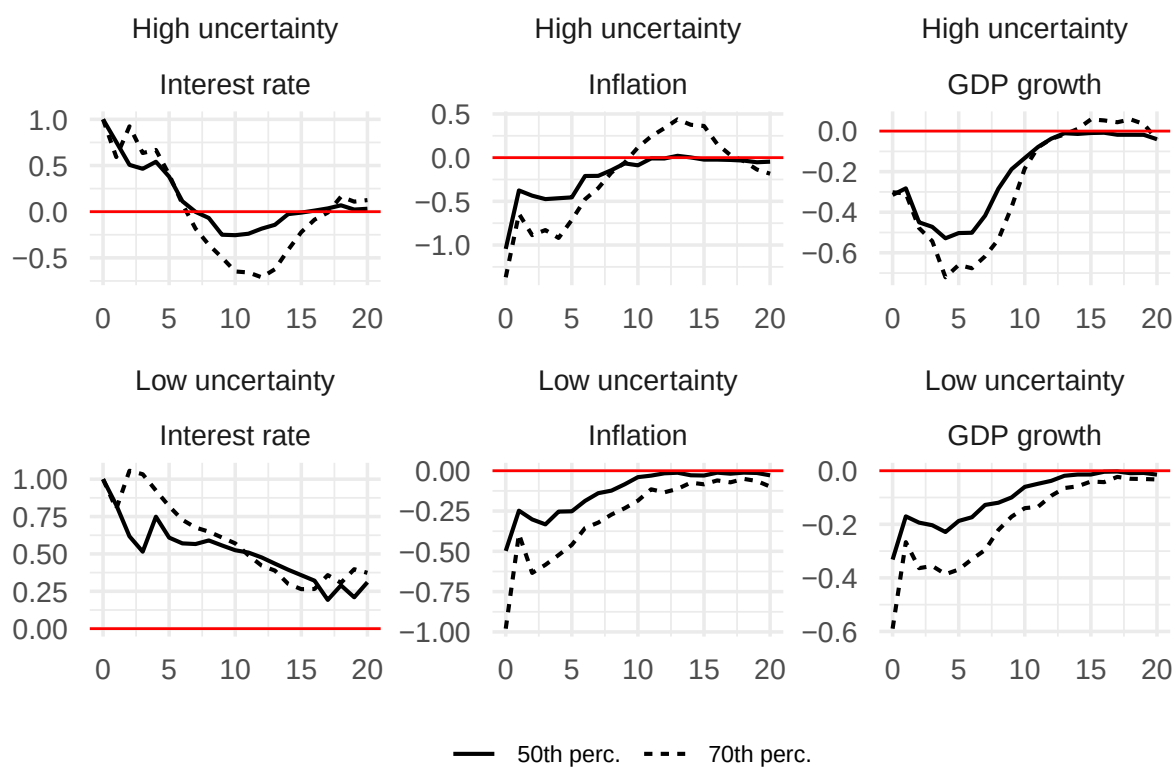


Figure 15: Global IRFs with the EPU, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in the whole sample with valid impulse responses. The Economic Policy Uncertainty Index is used as the threshold variable.

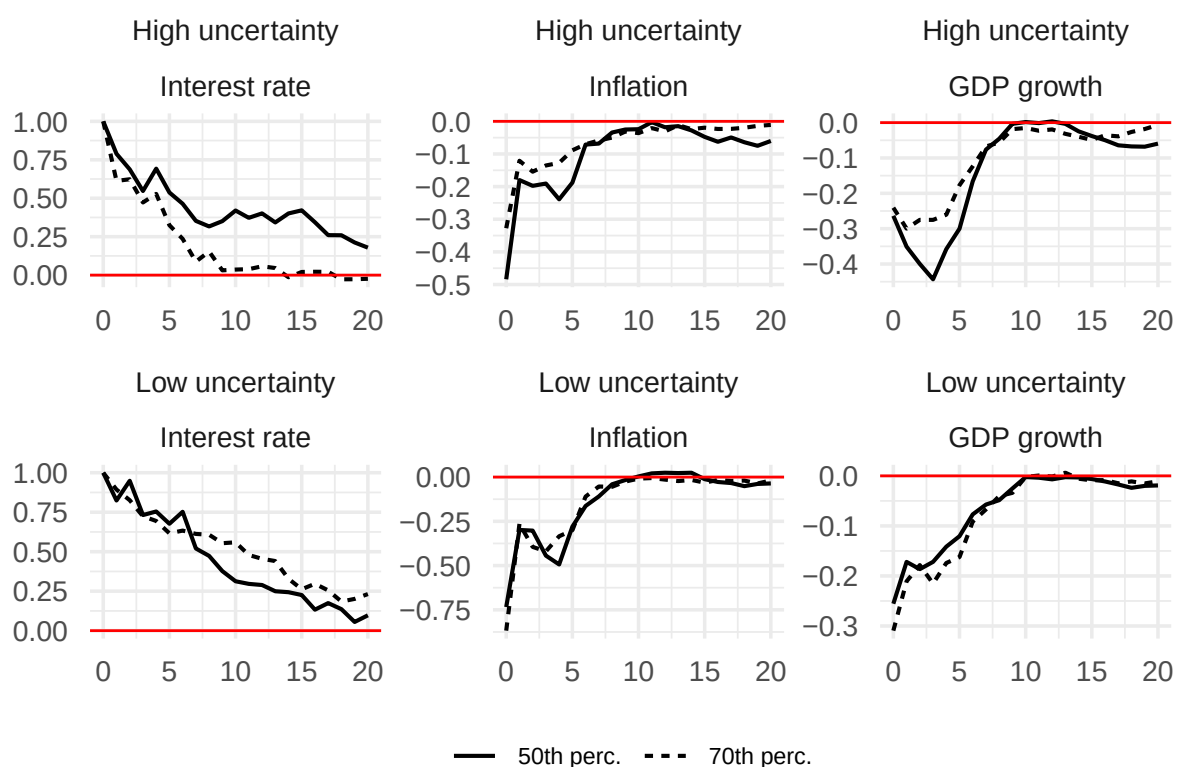


Figure 16: Global IRFs with the GPR, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in the whole sample with valid impulse responses. The Geopolitical Risk Index is used as the threshold variable.

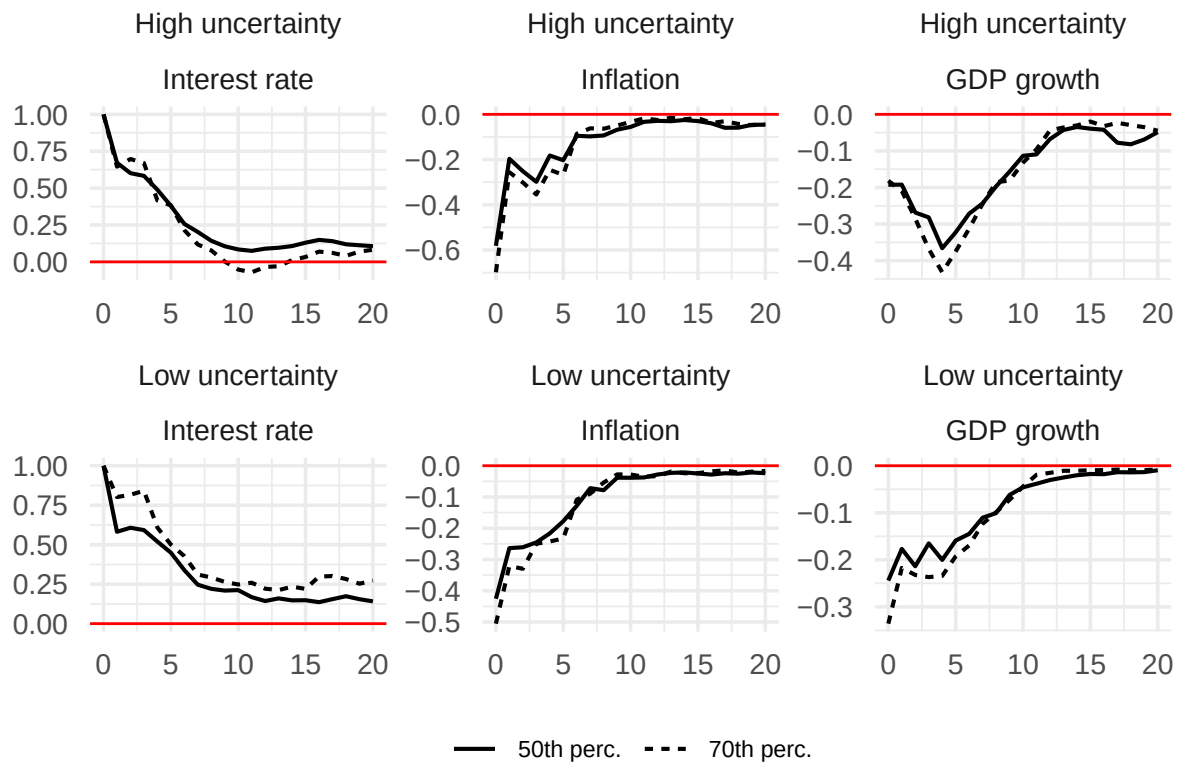


Figure 17: Global IRFs with the TPU, median and 70th perc. thresholds.

Notes: Point-by-point median impulse responses comparing threshold levels at the 50th and 70th percentiles in the whole sample with valid impulse responses. The Trade Policy Uncertainty Index is used as the threshold variable.